



RESEARCH ARTICLE

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Identifying MESSENGER Magnetospheric Boundary Crossings Using a Random Forest Region Classifier

Key Points:

- A random forest was used to classify magnetospheric regions based on MESSENGER magnetometer and ephemeris data
- The model had a high accuracy on unseen data, with a score of 0.9853 ± 0.0002
- 19,989 bow shock crossings and 11,414 magnetopause crossings were determined for the MESSENGER mission

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract We present a new list of bow shock and magnetopause crossings based on automated region classification for the MESSENGER (Mercury Surface, Space Environment, Geochemistry and Ranging) mission. We fit a random forest model to magnetometer and ephemeris data to classify the solar wind, magnetosheath, and magnetosphere regions surrounding Mercury. The random forest was highly accurate when predicting the testing data set, with an accuracy of 0.9853 ± 0.0002 . We apply this model to the orbital phase (March 2011 to April 2015) of the MESSENGER mission, and determine crossings automatically where changes in region classification occur. We apply several post-processing steps to this list of crossings and determine 19,989 bow shock crossings and 11,414 magnetopause crossings for the MESSENGER mission. We find good visual comparison with the spatial distribution of these new crossings and existing data sets of boundary crossing intervals.

Plain Language Summary Surrounding the planet Mercury is a large and dynamic magnetic bubble (known as a magnetosphere), which changes in size and shape due to interactions between Mercury's magnetic field and the solar wind (a stream of charged particles from the Sun). Due to changing external conditions, the location of the boundaries of this bubble can vary rapidly and dramatically. NASA's MESSENGER spacecraft spent 4 years orbiting Mercury between 2011 and 2015, and sampled from different regions of space with respect to this bubble which have distinct physical differences. We use a simple, yet powerful, machine learning technique to build a model to identify these regions, to ultimately determine a list of times when the spacecraft crossed the boundaries between these regions.

1. Introduction

Mercury has the weakest intrinsic magnetic field of all the planets in the Solar System. This, coupled with its close proximity to the Sun, produces extreme and highly variable magnetospheric interactions (Milillo et al., 2020). Among other effects, variations in the upstream solar wind conditions cause highly dynamic fluctuations in the morphology of its magnetospheric boundaries, the bow shock and the magnetopause. The bow shock marks the discontinuity where the solar wind is rapidly decelerated, compressed and deflected as Mercury's magnetosphere acts as an obstacle in its path (Milillo et al., 2020). The magnetopause lies at the equilibrium between the internal magnetic pressure of the magnetosphere and the external solar wind pressure (Shue & Song, 2002). Relative to other magnetized planets in the solar system, Mercury has the smallest magnetosphere, with an average magnetopause standoff distance of only 1.45 Mercury radii ($R_M = 2,438$ km) (Winslow et al., 2013). Measuring the locations of these boundaries is important for understanding the timescales and magnitudes of magnetospheric compressions (Slavin et al., 2019), separating plasma populations to enable large scale studies (Bowers et al., 2024; Karlsson et al., 2024), and informing models (He et al., 2022; Philpott et al., 2020; Winslow et al., 2013; Zhong et al., 2015).

Three spacecraft have previously sampled Mercury's boundaries. Mariner 10 was the first to encounter Mercury's bow shock and magnetopause across three flybys between 1974 and 1975 (Ness et al., 1974). MESSENGER (Mercury Surface, Space Environment, Geochemistry and Ranging) (Leary et al., 2007), launched in 2004, made three Mercury flybys (two in 2008 and one in 2009 (Baker et al., 2011; Slavin et al., 2008)) before entering continual orbit in 2011. This was followed by almost 4 years of exploration, with orbits taking the spacecraft within the vicinity of the boundaries thousands of times. BepiColombo is a two spacecraft mission on the way to Mercury, which has undergone 6 flybys to date (Benkhoff et al., 2021), and Mercury orbit insertion is expected in late 2026.

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Crossings of the bow shock and magnetopause are visually identifiable in data from spacecraft magnetometers as well as other instruments such as plasma, energetic particle and wave sensors. Bow shock crossings are typically identified by large changes in the magnitude or variability of the magnetic field (Jackman et al., 2019; Kruparova et al., 2019; Winslow et al., 2013). Bow shock crossings can be split into two broad categories: quasi-perpendicular and quasi-parallel, based on the angle between interplanetary magnetic field (IMF) vector and the bow shock surface normal. Quasi-perpendicular crossings are typically identified through rapid changes in the magnitude of the magnetic field. They also often include a shock foot and overshoot (Dimmock et al., 2019). Quasi-parallel crossings are instead typically identified through changes in variability (Nevskii et al., 2023). Back-streaming particles create turbulent conditions around the shocked front which result in increased variability in the magnetic field (Jarvinen et al., 2020; Milillo et al., 2020). For these reasons, quasi-parallel shocks are often more difficult to visually determine.

Magnetopause crossings are typically identified through changes in magnetic field variability or through rotations of the magnetic field. The magnetosheath is often turbulent and compressed, producing a highly variable magnetic field in magnitude while the magnetosphere is much less variable (Haaland et al., 2014). Furthermore, the magnetic field orientation within the magnetosheath depends on the IMF orientation, while the magnetic field orientation within the magnetosphere is relatively steady as the direction of the field is determined by the planetary dipole. Magnetopause crossings are hence most easily identified when the magnetosheath magnetic field vector and magnetosphere magnetic field vector are separated by a large angle, or when there is a large change in variability of the magnetic field (Winslow et al., 2013). Due to the extreme variability of Mercury's magnetospheric boundaries (Milillo et al., 2020), as a spacecraft passes across a boundary, we often observe multiple crossings as the boundary moves back and forth over the spacecraft. Because of this, crossings at Mercury have previously been considered instead as intervals of time in which multiple crossings occur, rather than a series of individually identified boundary crossings. Intervals of crossings of Mercury's bow shock and magnetopause have been visually identified with the above methods by Winslow et al. (2013); Philpott et al. (2020); Sun et al. (2020) and Sun (2023) for the duration of the mission. Further detail on this labeling can be found in Section 2.1.

Automated boundary detection methods, including machine learning techniques, were explored previously at Mercury and other planets, discussed further in Section 3.1. Automated crossing detection was previously achieved through both the direct detection of changes within data (for example, Daigavane et al., 2022; Nevskii et al., 2023), but also through indirect methods, where crossings were determined based on changes in region classification (for example, Bakrania et al., 2020; Olshevsky et al., 2021; Wang et al., 2025). Yet, no automated method currently exists to determine individual crossings of Mercury's boundaries. The magnetospheric region in which a spacecraft is at a given time is often trivial to determine visually for single event case studies by those with domain knowledge. However, over longer timescales automated identification offers a quicker approach. Furthermore, visual-based selection may also contain inherent biases potentially producing an inconsistent result. In this work, we present a region classifier model based on numerical parameters of the data, resulting in an unambiguous and reproducible identification of boundary crossings. This method could easily be rolled out onto future unseen data without significant additional effort.

In this manuscript, Section 2 outlines the MESSENGER mission, its available data and their applicability to boundary identification. Section 3 describes the methods used to model region classification, along with the application of that model. Section 4 details the results of model testing and the resulting list of boundary crossings along with example applications of the model. Section 5 further explores the new crossing list and examines spatial distributions and trends in heliocentric distance. This section also outlines avenues of future research.

2. Data

MESSENGER orbited Mercury from March 2011 to April 2015, after which it was de-orbited (Milillo et al., 2020). Its instrumentation included a magnetometer (MAG) which sampled the magnetic field at a cadence of up to 20 Hz (Anderson et al., 2007; Leary et al., 2007). MESSENGER initially orbited Mercury with a period of 12 hr, however, after April 2012 it was shortened to 8 hr Figure 1 shows examples of MESSENGER's orbit in the aberrated Mercury Solar Magnetic (MSM') coordinate system, where the origin is centered at Mercury's magnetic dipole (offset ~479 km northward of the planet's center (Anderson et al., 2011; Johnson et al., 2012)), $X_{\text{MSM}'}$ points into the flow of solar wind (aberrated from sunwards due to Mercury's orbital velocity—which is

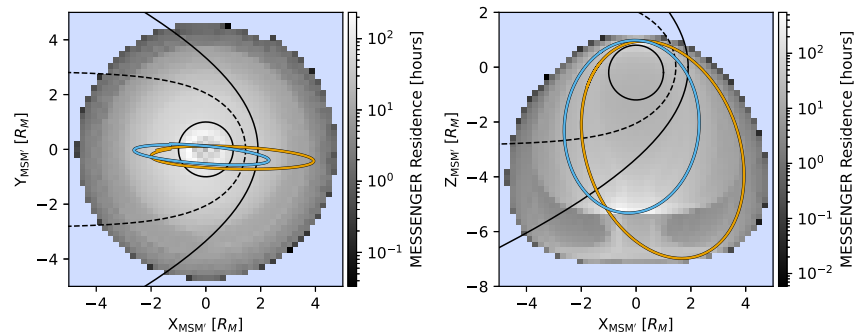


Figure 1. Two example orbits of the MESSENGER mission plotted in the XY and XZ planes of the aberrated Mercury-Solar-Magnetic (MSM') coordinate system. The background histogram denotes the residence time of MESSENGER as projected to a given spatial bin in each plane. A pale blue background denotes regions where MESSENGER did not sample. A circle is drawn to represent the size and location of Mercury. The average boundary shape for the bow shock (solid curve) and magnetopause (dashed curve) as determined by Winslow et al. (2013) are included for context. The orange line plots the orbit of MESSENGER starting 25 December 2011 02:00, the blue line plots the orbit of MESSENGER starting 5 June 2013 01:00.

significant with respect to the solar wind velocity). $Z_{MSM'}$ points northward along the rotation axis, and $Y_{MSM'}$ completes the right-handed orthogonal set. In our work, we follow methods employed in previous works and calculate this aberration angle daily, assuming an average solar wind velocity of 400 km s^{-1} (Bowers et al., 2024). The aberration angle varies between 5.6° at aphelion and 8.4° at perihelion. MESSENGER's periaapsis stays close to the surface north of the planet, while its apoapsis extends approximately $7R_M$ south of the planet before shortening to $5R_M$. This highly elliptical orbit enables the spacecraft to sample from the magnetosphere, magnetosheath, and solar wind, all in one orbit. In Figure 1, the orange line shows an orbit with a period of 12 hr. The orbit is aligned to the XZ plane, and tilted sunward (positive X), meaning it spends a large portion of the orbit within the solar wind. In contrast, the blue line shows an orbit tilted anti-sunward (toward negative X), and spends a smaller portion of the orbit within the solar wind. This is furthered by this orbit having a lower apoapsis, and hence a shorter period. The background of Figure 1 shows a 2D histogram of MESSENGER's residence as projected to each plane. The edges of MESSENGER's extent are darker and therefore less sampled, while bins closer to the planet were sampled more frequently.

2.1. Instrumentation

We utilize the MESSENGER MAG Calibrated Data Bundle (Anderson et al., 2007; Korth & Anderson, 2016) during the orbit phase of the MESSENGER mission. This data set consists of the time, vector magnetic field, and spacecraft position for the duration of the mission. Measurements were recorded with a maximum sample rate of 20 Hz (downsampled averages were also made available at 1, 5, 10, and 60 s intervals).

Onboard MESSENGER was also the Fast Imaging Plasma Spectrometer (FIPS) instrument which measured the energy, angular distributions, and compositional distributions of charged particles between 46 eV q^{-1} and 13 keV q^{-1} (Andrews et al., 2007). Generally, information about the plasma in a magnetospheric environment is extremely useful in categorizing spacecraft location with respect to that environment. Many previous efforts to classify spacecraft location with respect to magnetospheric region at Mercury and other planets in the solar system have utilized plasma spectra and derived products in their methods: Bakrania et al. (2020); James et al. (2020); Daigavane et al. (2022); Grison et al. (2025). However, FIPS data have several constraints which make them less useful for this work in particular. The nominal measurement cadence of FIPS of 10 s (burst mode, near planet), and 65 s (normal mode) (Andrews et al., 2007), limits the scale at which classifications based on these data can be made. Through visual inspection of MAG data we find individual crossings can occur just seconds apart. To classify changes in region on this short of a timescale, we require a measurement cadence on a similar or shorter timescale. FIPS can also run in a faster mode, with a 2 s measurement cadence (Andrews et al., 2007), however this was not consistently in use. Secondly, a large part of FIPS's field of view was occluded by MESSENGER's sunshade (Andrews et al., 2007), restricting the distribution of plasma measured as sources from the sunward direction were obstructed, making the measurements less reliable. For these two reasons, we choose to not use FIPS plasma spectra or derived products in this work.

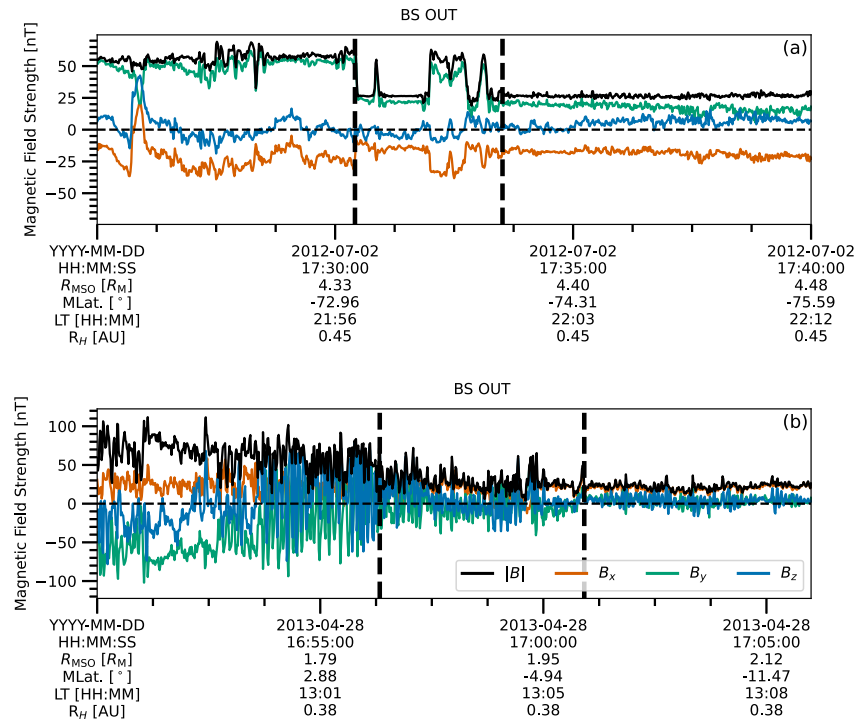


Figure 2. Contrasting examples of magnetometer data from MESSENGER for two outbound bow shock crossing intervals as labeled by Philpott et al. (2020). Vertical dashed lines mark these intervals, along with the label BS OUT. Panels (a, b) show examples of quasi-perpendicular and quasi-parallel cases respectively.

2.2. Previous Labeling of Boundary Crossing Intervals

We based this study on previous labeling of the MESSENGER MAG data set by Philpott et al. (2020) which further expanded labeling by Winslow et al. (2013) to the entire mission. They identified intervals of time grouping sequences of crossings of the same magnetospheric boundary together. These intervals were labeled by boundary (BS: bow shock, MP: magnetopause) and direction (IN: inbound—toward the magnetosphere, OUT: outbound—away from the magnetosphere), for example BS_IN, MP_OUT. These intervals were visually determined based on manual inspection of the MAG data alone, and marked sections of time which span the first and last crossing in a given pass through the boundary. Though multiple lists of boundary crossing intervals are available (for example, Philpott et al., 2020; Sun, 2023), for this study we focus on the list by Philpott et al. (2020), which was cross-validated by members of the MESSENGER MAG science team. This list labels 16,266 intervals: 8,108 for the bow shock, 8,131 for the magnetopause, and 27 data gaps.

Examples of these intervals within the context of the data are shown in Figures 2 and 3 for bow shock and magnetopause intervals respectively. Figure 2a shows a typical case for a quasi-perpendicular crossing of Mercury's bow shock. We observe two distinct regions in the MAG data: high amplitude magnetosheath before the crossing interval and low magnitude solar wind after the crossing interval. Figure 2b instead shows a typical case for a quasi-parallel crossing of Mercury's bow shock, characterized by significantly higher variability than the above case. In Figure 3a, the magnetopause crossing itself was visually distinct as there was a rotation in the field resulting in a large change in B_x across the boundary. Figure 3b instead shows an example where the field vector in the magnetosphere and magnetosheath are closely aligned. There was very little rotation across the boundary, and the exact point of crossing is difficult to discern. We show the spatial distribution of the Philpott et al. (2020) crossing intervals in Figure 4 as projected to the $XY_{MSM'}$ (a, d), $XZ_{MSM'}$ (b, e), and cylindrical (c, f) planes. Panels (c) and (f) show the spatial distribution matching well to the average boundary location by Winslow et al. (2013). This was to be expected, as this boundary shape was determined using a subset of these intervals. Due to the geometric constraints of Mercury's orbit (as described in Figure 1) crossings were predominantly observed in negative $Z_{MSM'}$ and never northward of $Z_{MSM'} = 1.5$. Crossings were observed at all local times—with a skew toward noon.

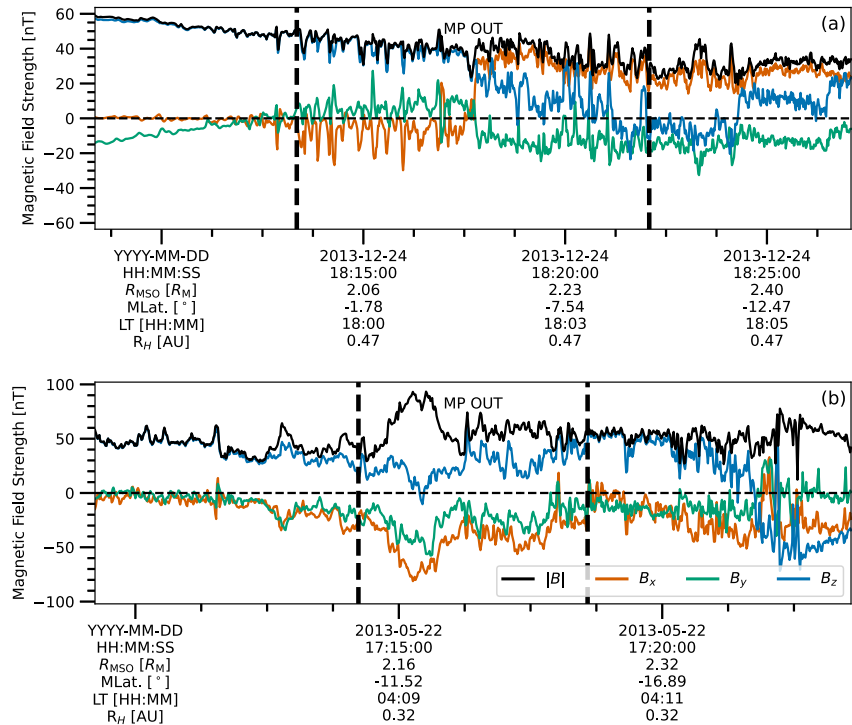


Figure 3. Contrasting examples of magnetometer data from MESSENGER for two outbound magnetopause crossing intervals as labeled by Philpott et al. (2020). Vertical dashed lines mark these intervals, along with the label MP OUT. Panel (a) shows an example containing a clear and singular rotation of the field orientation (at approximately 18:18 on 22nd May 2013), while panel (b) shows a less clear example.

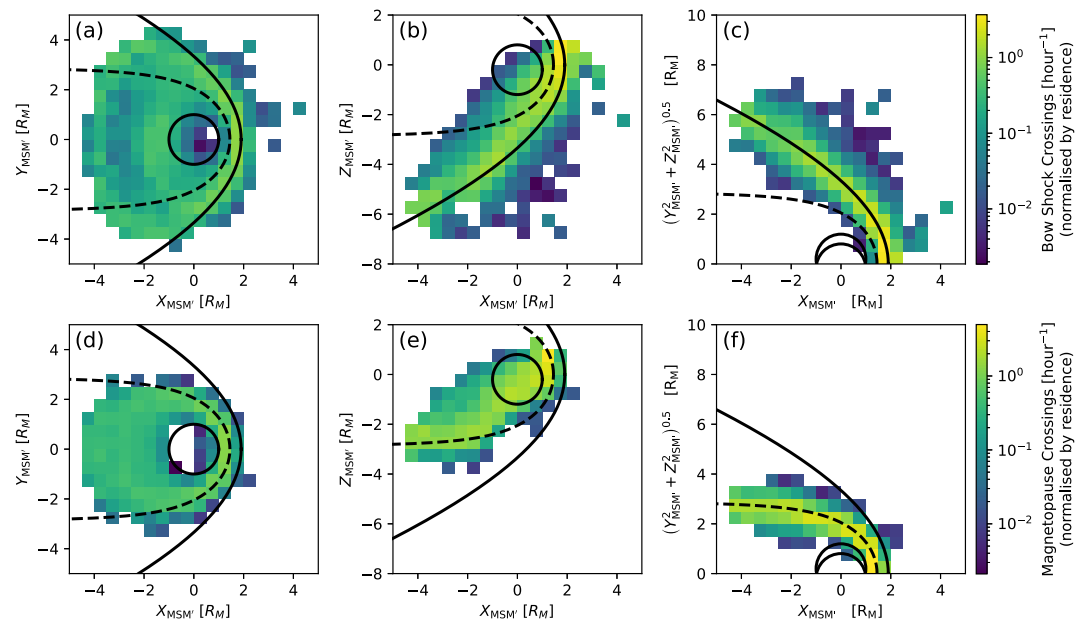


Figure 4. The spatial distribution of bow shock (a–c) and magnetopause (d–f) crossing intervals from Philpott et al. (2020) are shown as projected to the XY and XZ planes of the aberrated Mercury-Solar-Magnetospheric (MSM') coordinate system. The density of crossing intervals is normalized by the time spent in a given spatial bin, resulting in a crossing rate in units of crossings per hour. A circle is drawn to represent the size and location of Mercury. The average boundary shape for the bow shock (solid curve) and magnetopause (dashed curve) by Winslow et al. (2013) (based on these data) are included for context.

While useful for visualizing the system, it is somewhat arbitrary to describe an interval of time with a single position. As shown in Figure 2a, bow shock and magnetopause crossing intervals often contain more than one individual crossing. By treating multiple crossings as one entity, we lose relevant statistical information about the boundaries, such as the exact locations/times and the temporal density (how closely spaced they are) of individual crossings—and hence the dynamic temporal variability of Mercury's magnetosphere. The aim of this work was to build on this list of boundary crossing intervals by utilizing them to first isolate regions of pure solar wind, magnetosheath, and magnetosphere within MESSENGER MAG data and ephemeris data. Based on this region classification, we constructed a data set upon which to train a machine learning model, to automatically identify every individual boundary crossing within each crossing interval.

2.3. Data Preparation

We opted to use a supervised machine learning technique to first classify magnetospheric region (i.e., solar wind, magnetosheath, magnetosphere) at Mercury, and only then determine spacecraft crossings of the boundaries based on changes in that region classification. The MESSENGER MAG data set records 6 covariates from each time step: the vector magnetic field, (B_x, B_y, B_z) , and the vector position $(X_{\text{MSO}}, Y_{\text{MSO}}, Z_{\text{MSO}})$, both in MSO coordinates—which we converted to MSM' by rotating by the aberration angle, and offsetting the position by the dipole offset described at the start of Section 2. To create a labeled data set of magnetospheric regions, we treated each MESSENGER observation in time as an independent random sample, but incorporated some time-dependency information by deriving features from 10-s long overlapping windows, each spaced 1 s apart. The length of this window was chosen as a compromise between minimizing variation due to changing conditions within the window, and maximizing the length to ensure the data were representative of the overall region distribution. We derived statistical features from the distribution of data within each 10-s window of data. For ephemeris features (any feature derived from the position of the spacecraft), we only recorded the position at the mid-point of the window as a feature for that window. In total, we derived 28 features from these data: the mean, median, standard deviation, skew, and kurtosis of the magnitude and components of the magnetic field, the magnitude and components of MESSENGER's position in MSM' coordinates and further metrics of MESSENGER's position relative to Mercury: latitude, magnetic latitude, local time, and heliocentric distance. Figure 5 shows an example of two such prepared samples, their distribution of magnetic field data, and the derived features.

We constructed a labeled data set of 10-s windows with the following method:

1. Iterating through the list of crossing intervals by Philpott et al. (2020), we loaded data within 10 min before and after each interval. These “interval extensions” delineated the regions within which we selected training data. Figure 6 shows an example of these extensions surrounding an outbound bow shock crossing interval. In panel (a), the orange and yellow regions denote the two interval extensions corresponding to magnetosheath and solar wind training data respectively. Panels (b) and (c) highlight how the distributions of magnetic field data are distinct between the two regions. We limited these selection regions to 10 min to avoid large overlap between successive interval extensions.
2. We then checked for overlap between interval extensions and the neighboring intervals. For some crossing intervals, primarily those on the dayside, the length of time between successive bow shock and magnetopause crossing intervals can be less than 10 min. As a result, the 10 min interval extension following one crossing interval could overlap with the start of the next interval, inadvertently including and mislabeling data from an interval in the training data. These cases occur for approximately 13% of intervals. To prevent this, we limited the length of the extensions to avoid any overlap with crossing intervals such that all extensions were ≤ 10 minutes.
3. Within each interval extension, we randomly selected 10 of these 10-s windows to add to the training data set. We selected only 10 samples from each extension to reduce computational time, however, selecting multiple samples from each interval was critical to avoid potential anomalies in the data, such as the brief rotation of the magnetic field on the 28 November 2011 10:44 in Figure 6.
4. For each sample, we calculated the features as described and shown in Figure 5, and labeled the sample's region based on the label of its reference crossing interval. for example if the sample was taken after a BS_IN interval, it was labeled as magnetosheath.

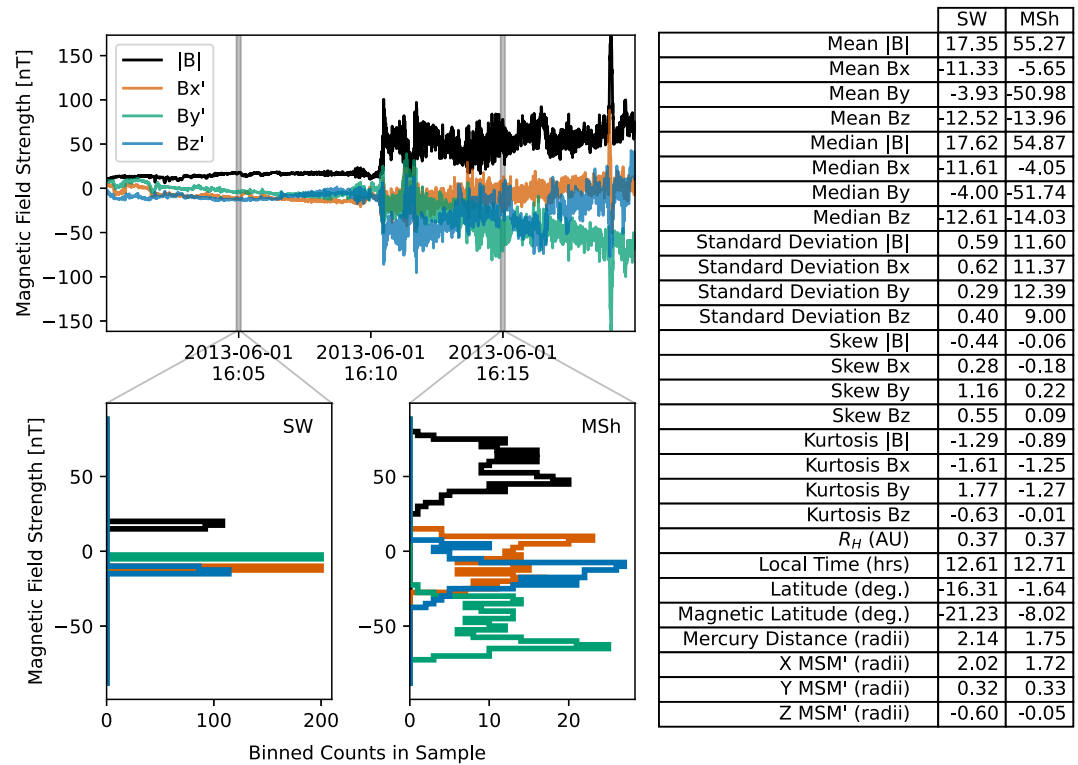


Figure 5. An example of how features are extracted from a given sample of data. Panel (a) shows an example of MESSENGER MAG data before and after a sequence of inbound bow shock crossings. Two 10 s long samples are arbitrarily selected from before and after this crossing—shaded in gray. Below, in panels (b, solar wind) and (c, magnetosheath) we show the distributions of magnetic field data in each of these samples. Panel (d) shows the numerical value of each feature derived from these samples.

As a result of this method of construction, this data set contained a class imbalance. There were only two interval types which yielded samples of solar wind (BS_IN and BS_OUT) and of magnetosphere (MP_IN and MP_OUT), however, all four interval types yielded magnetosheath samples. This resulted in twice as many magnetosheath samples as solar wind or magnetosphere. In addition to this, we identified and removed extreme (non-physical) values from the training data set by stipulating that mean and standard deviation of $|B|$ were less than 5,000 nT. We chose this value, based on visual inspection of the training data set, to exclude obvious outliers. Following the removal of extreme samples, we randomly resampled all classes to match the minority class size. This left us with a data set containing 243,234 samples, 81,078 in each class. The distribution of features in this data set can be found in Text S1 of Supporting Information S1. We randomly split these data into a training and a testing data set, with a ratio of 9:1. When later applying the model to the unseen data (i.e., within the crossing intervals), the same data preparation technique was applied.

3. Methodology

3.1. Machine Learning Approach

Multiple approaches to the automatic identification of individual crossings have been explored in previous works for many planets in the Solar System. Individual crossings have previously been identified directly through the use of change point detection (Daigavane et al., 2022; Nevskii et al., 2023), or based on the thresholding of data (Case & Wild, 2013). Indirect methods are also viable, where magnetospheric regions (e.g., solar wind, magnetosheath, magnetosphere) are first classified, and individual crossings are then placed where these regions change. Previously, this has included decision tree methods (Nguyen et al., 2022; Wang et al., 2025), unsupervised methods (Bakrania et al., 2020; Edmond et al., 2024; Toy-Edens et al., 2024), and neural networks (Argall et al., 2020; Breuillard et al., 2020; Julka et al., 2023; Kit Cheng & Achilleos, 2022; Olshevsky et al., 2021). Each of these studies was tailored to the nature of the data sets used and spacecraft available.

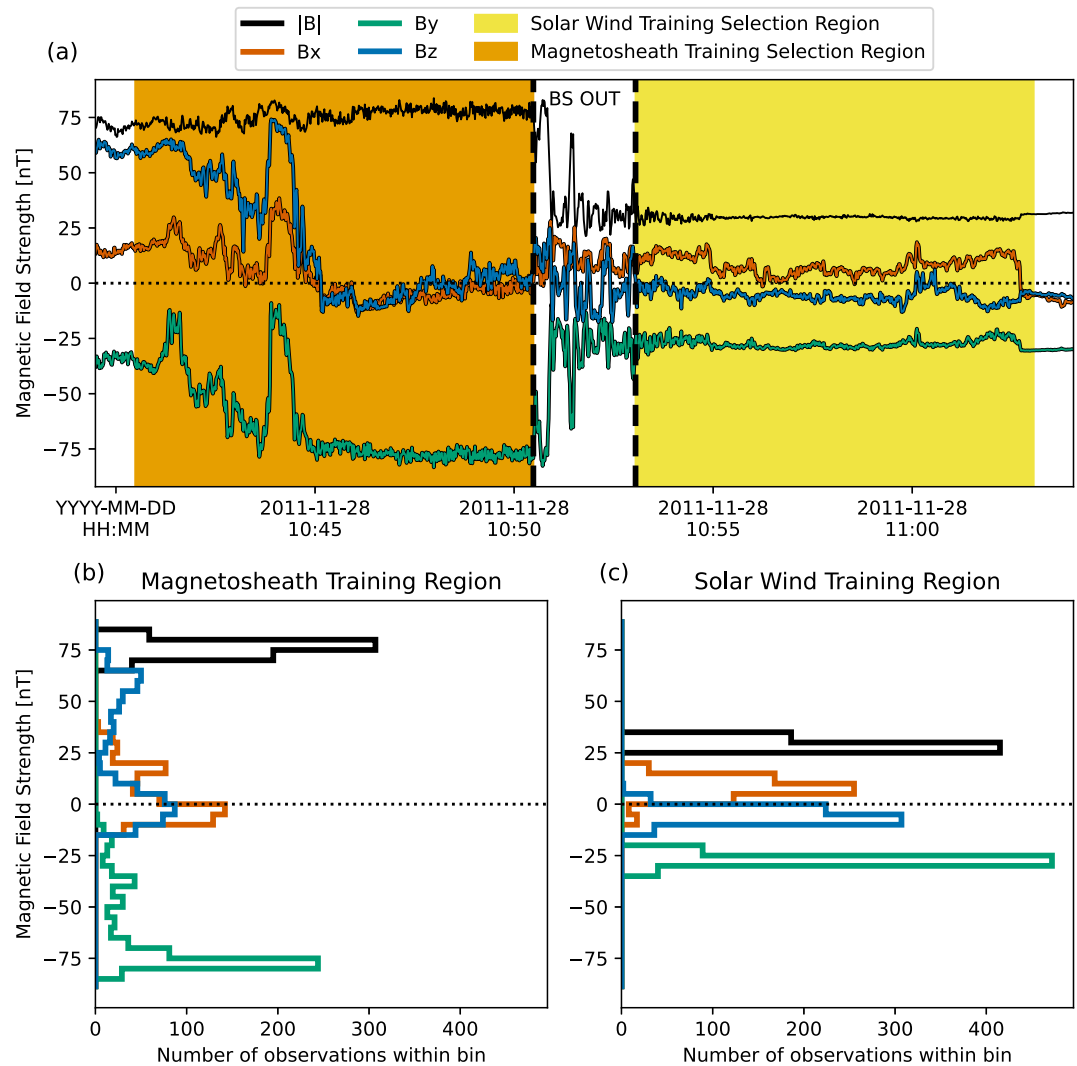


Figure 6. Example MAG data from MESSENGER during and surrounding an outbound bow shock crossing. Panel (a) shows the three-component magnetic field vector in MSM' coordinates, along with the magnitude (black). Shaded regions mark 10-min-long sections of “pure” magnetosheath (left, orange) and solar wind (right, yellow) within which we sampled training data. Unshaded regions are not considered for training purposes. Panels (b, c) show the distributions of these data (binned every 5 nT) within each shaded region for the magnetosheath and solar wind respectively. From these longer intervals of time, ten, 10-second-long samples were drawn and added to the prepared data set.

We choose to use a random forest model (Breiman, 2001) in our work for their computational efficiency, flexibility of fit, robustness to outliers and noise, and insensitivity to overfitting (Belgiu & Drăguț, 2016; Breiman, 2001). Specifically, we use the `RANDOMFORESTCLASSIFIER` implementation from Scikit-learn, a Python machine learning toolkit (Pedregosa et al., 2011). Random forests are an unconventional tool for working with time series, as the time-dependent structure is not modeled. Instead, we treated each observation in time as an independent random sample (as described in Section 2.3). We evaluated 6 different feature-sets to use in modeling. For each feature-set, we trained 10 random forest models with default hyperparameters on all of the training data and recorded training accuracy and out-of-bag (OOB) score (each with different but fixed random seed—initialized at 1785, and incremented by 1 where variability was required. For more details, see the implementation in Hollman et al. (2025a)). Mean and standard deviation of the training accuracy, and the OOB Score are included in Table 1. “All Features” is a feature-set containing all of the features described above. “Reduced Features” removes skew and kurtosis features. “No Ephemeris” removes all positional features of the spacecraft, such as X , Y , and Z MSM', local time, latitude, etc. “Only Mean”, “Only Median”, and “Only Standard Deviation” contain only 4 features each, for

Table 1
Evaluation Metrics for 6 Different Feature-Sets

Feature set	Training accuracy		OOB score	
	Mean	SD	Mean	SD
All Features	~ 1	1×10^{-6}	0.9838	0.0001
Reduced Features	~ 1	1×10^{-6}	0.986	0.0001
No Ephemeris	0.999997	2×10^{-6}	0.9145	0.0002
Only Mean	0.99998	1×10^{-5}	0.8297	0.0002
Only Median	0.999981	6×10^{-6}	0.825	0.0003
Only Standard Deviation	0.999984	7×10^{-6}	0.6751	0.0004

Note. The mean and standard deviation training accuracy and out-of-bag score are shown for 10 model runs, each with different but fixed random seed.

the mean, median, and standard deviation of the field magnitude and components respectively. We found the “Reduced Features” to be most optimal, with a mean OOB score of 0.986 ± 0.0001 .

We used the Python package OPTUNA to vary a selection of model hyperparameters during model training (Akiba et al., 2019). Using the optimal feature-set found above, we trained 100 random forest models on the prepared training data set, allowing hyperparameters to vary while optimizing for minimum OOB error. Each model had a different random seed (each fixed to ensure reproducibility). We recorded the hyperparameters which were varied, and their optimum values in Table 2. To measure the variability of prediction accuracy for this final model architecture, we trained 30 further instances of this model (with different but fixed random seed) with the selected optimal feature set hyperparameter settings, evaluating their performance on the testing data set. We saved the model with the largest accuracy score for further application to unlabeled data within the crossing intervals.

3.2. Model Application

Due to the regularity of MESSENGER's orbit, with periapsis deep within the magnetosphere and apoapsis far out in the solar wind, combined with the pre-labeled boundary intervals, we saved on computational requirements by restricting our application to the crossing intervals themselves along with the surrounding 10 min on either side. However, we have included an example classification to a full orbit of MESSENGER in Text S2 of Supporting Information S1. While we treated the crossing intervals by Philpott et al. (2020) as ground truth for the purposes of training, we do not want to rule out the possibility of mislabeling in the intervals data set. In application of the model, we grouped crossing intervals into pairs where possible to ensure a consistent classification between intervals. This consisted of pairing consecutive bow shock and magnetopause intervals (i.e., BS_IN followed by MP_IN, or MP_OUT followed by BS_OUT), the average time between intervals being ~ 42 minutes. Not all boundary crossing intervals exist in pairs however, due to the existence of data gaps and a small number of cases where a boundary did not manifest clearly in the data (and hence could not be labeled). This can happen, for example, when a strong ICME hits the planet, disrupting the regular bow shock and magnetopause structure and reconfiguring the environment spatially (for example, Bowers et al., 2025).

Iterating through these groups of crossing intervals (either paired or unpaired), we took the following steps to apply the model:

1. First, the full cadence MESSENGER MAG time series was loaded from 10 min before the first crossing interval of the pair, to 10 min after the second. When unpaired, data were loaded from 10 min before, to 10 min after the single crossing interval.
2. For each second of data loaded, we calculated features from the surrounding 10 s of data as described in Section 2.3.
3. We then passed these features to the model to determine both a class prediction, along with class probability for the three classes (with the class probability of a single tree being the fraction of samples of the same class in a leaf, and the class probability from the model being the average probability of all trees).

Table 2
A Subset of Hyperparameters Used by the Random Forest Model Which Were Optimized During Training

Hyperparameter	Value	Sampling bounds	Description
n_estimators	500	50, 500	The number of trees in the forest
max_depth	56	10, 100	The maximum depth of the trees
max_features	$\sqrt{N_{\text{features}}}$	Choice of $\log_2 N$, $\sqrt{N}$, or None	The number of features to consider for a given decision tree
min_samples_split	2	2, 10	The minimum number of samples required to split a node

Note. Further detail, and a full list of parameters, can be found in the Scikit-Learn documentation for `RANDOMFORESTCLASSIFIER`.

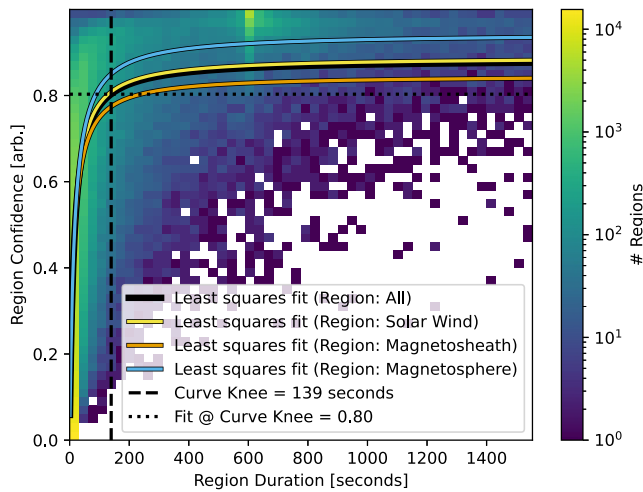


Figure 7. A 2D histogram of region confidence and duration for all regions, determined based on the raw model outputs. A solid black line shows a least squares regression with the functional form $f(x) = \frac{Ax}{B+x}$ ($A = 0.883 \pm 0.001$, $B = 13.99 \pm 0.04$). Horizontal and vertical black dashed lines intersect the curve at the knee point—determined with the *kneedle* method (Satopaa et al., 2011). Yellow, orange, and blue lines show this function as fit to only regions classified as solar wind, magnetosheath, or magnetosphere respectively (with parameters: $A_{SW} = 0.89 \pm 0.001$, $B_{SW} = 13.7 \pm 0.07$, $A_{MSh} = 0.846 \pm 0.001$, $B_{MSh} = 13.13 \pm 0.05$, $A_{MSP} = 0.942 \pm 0.002$, $B_{MSP} = 15.21 \pm 0.09$).

After iterating through each group of crossing intervals and applying the model with the above steps, the classification times, and class probabilities were saved together—referred herein as the “raw model output.”

3.3. Determining Boundary Crossings

To place crossings based on the three class probabilities, we first split the raw model output back up into the groups of paired crossing intervals (as described in Section 3.2). For each of these groups, we determined the classified region by the largest class probability from the model. We then placed crossings where this classification changed, halfway between the two surrounding classifications. When classifications were uncertain—meaning there was no distinct classification, that is the ratio of the second highest class probability over the highest class probability was close to 1—the variation in subsequent classifications resulted in many crossings in short succession. This was unwanted behavior as we lacked confidence in the region classifications which dictate those crossings, and hence could not be certain that these were physical crossings. To handle this, we first combine successive region classification of the same region into a contiguous region describing a longer time. For each of these contiguous regions, we determine an associated “region confidence” as the complement of the mean of the aforementioned probability ratio within the region. For the first and last region of any application, we assume a region confidence of 1. To mark uncertain regions as unknown, we introduce the following criterion. Figure 7 shows a 2D histogram of region confidence against region duration for all regions classified (with outliers in duration, $>3\sigma$, $\sim 1.7\%$, removed).

We applied a non-linear least squares regression from the *SCIPY* Python package (Virtanen et al., 2020) of functional form $f(x) = \frac{Ax}{B+x}$ to parametrize

a curve describing the average relationship between region confidence and duration. This function was chosen as its form matched our visual observations of the data: a sharp decrease in average confidence at short durations, and converging to a constant average confidence at large durations. Based on the knee-point of this curve (calculated using the *kneedle* method (Satopaa et al., 2011)—determined by the maximum vertical distance of the curve from a straight line with the average slope of the curve), we imposed that any region shorter than the curve duration at the knee (139 s) would be re-classified as unknown, unless it was more confident than the curve confidence at the knee (0.8), that is removing the lower left quadrant in Figure 7. We choose this metric (rather than rejecting all regions under the curve) as we want to keep long-duration regions—even those with lower than average confidence. While the average confidence across the region is low, the continuous classification is indicative of a correct region classification. We can think of region duration as another metric of confidence. We additionally applied curves to this regions data set, subset by region type, however, we use the fit to all regions for determining unknown regions. An increase in columns near 600 s (10 min) corresponded to applications in which the first crossing was placed at the start of the crossing interval, yielding a region which spans from the start of the application to the start of the crossing interval, which by definition is 10 min. This could similarly occur from the end of the crossing interval to the end of the application.

The result was a list of contiguous region classifications of the following 4 classes: solar wind, magnetosheath, magnetosphere, and unknown. We hence defined 12 region transitions under 3 broad categories:

1. Direct crossings: Bow shock inbound and outbound (BS_IN, BS_OUT), and magnetopause inbound and outbound (MP_IN, MP_OUT).
2. Unknown crossing types for unknown transitions between the three regions (SW $\rightarrow$ UKN, UKN $\rightarrow$ SW, MSh $\rightarrow$ UKN, UKN $\rightarrow$ MSh, MSp $\rightarrow$ UKN, UKN $\rightarrow$ MSp).
3. Unphysical crossing types (SW $\rightarrow$ MSp, or MSp $\rightarrow$ SW) where the intermediate region is skipped. While this violates the nominal magnetospheric geometry, crossings classed as unphysical are not necessarily incorrect in all cases, as magnetospheric geometry can change under extreme conditions (for example, Bowers et al., 2025). Table 3 records the number of crossings grouped by these types.

Table 3

Raw Crossing Counts as Determined by Region Transitions in the Initial Model Output, Following the Determination of Unknown Regions, Along With Updated Crossing Counts After the Post-Processing Steps Described in Section 3.4

Crossing type	Counts	Post-processed counts
Total	100,901	31,408
BS_IN	6,953	9,998
BS_OUT	6,993	9,991
ALL BS	13,946	19,989
MP_IN	2,815	5,718
MP_OUT	2,791	5,696
ALL MP	5,606	11,414
UKN (SW → UKN)	11,173	
UKN (UKN → SW)	11,109	
UKN (MSh → UKN)	23,469	
UKN (UKN → MSh)	23,538	
UKN (MSp → UKN)	6,032	
UKN (UKN → MSp)	6,027	
UNPHYS (SW → MSp)	0	2
UNPHYS (MSp → SW)	1	3

Examples of applications of this method, including the determination of unknown regions and labeling of crossings are shown in Figures 8 and 9. Figure 8 shows an example classification with resulting crossing placement. In this example, we find three possible types of region transition. On the 3 April 2011 at 16:02 the model describes a transition from magnetosphere to magnetosheath through a short unknown region. This results in two unknown crossings labeled as UKN (MSp → UKN) for the magnetosphere to unknown region transition and UKN (UKN → MSh) for unknown to magnetosheath. Between the MP_OUT and BS_OUT intervals, this method described several unknown regions placed wholly within the same region (magnetosheath before and after). Unlike the first unknown region in this example, these don't necessitate a physical crossing as no confident change in region classification has occurred (and to eye there were no physical crossings for these). Unknown crossings were marked here as UKN (MSh → UKN) and UKN (UKN → MSh). Another example of this crossing type occurred within the BS_OUT interval. Lastly, the model placed several direct bow shock crossings, where transitions between magnetosheath and solar wind occurred directly, without an unknown region. There was one of these crossings at 17:30, marking brief motion of the boundary over and back across the spacecraft, before finally crossing into the solar wind at 17:43, with no more crossings occurring for this outbound passage. This example application highlights a relatively simple case for clarity of explanation, however in practice, more complex cases were common—an example of which is shown in Figure 9. This figure shows a quasi-parallel bow shock crossing interval, where the model application was uncertain (difference in class probabilities in Figure 9c was small), yielding frequent

changes in prediction. This example highlights how the introduction of unknown regions aids in removing false crossings which are due to fluctuations in uncertain predictions. Simpler cases also exist, with single crossing intervals also being commonplace.

In addition to this, there were also applications with unexpected results. An example of this is shown in Figure 10, where no crossings were predicted within the crossing interval itself. The model predicts magnetosphere before, during, and after the magnetopause crossing interval, and only predicts magnetosheath ~5 minutes after the end of the crossing interval. Furthermore, model probabilities (panel (c)) show that the change in prediction from magnetosphere to magnetosheath was gradual, indicating no distinct change in the data was present. We found approximately 5% of intervals didn't have a crossing predicted within 1 min of the interval, and approximately 1% didn't have a crossing predicted within 5 min. Similarly, there were applications where the unexpected class was classified within a crossing interval, that is solar wind classifications within a magnetopause crossing interval. There were 13 of these cases (<0.1% of intervals), all of which occur during magnetopause crossing intervals. No further processing steps were taken with respect to these cases as they were rare, and indicative of events of physical interest that warrant further study. We list the intervals in which these cases occur in Appendix B1.

3.4. Post-Processing Steps

We introduced several post-processing steps to address issues in the application of the model. When applying the model, data was only provided between fixed points, and any window querying data outside of this (i.e., those at the beginning and end of the application, <5 s from the edges of the application) contained missing values. As a result, many of these were incorrectly classified. To amend this, we remove the first and last 5 s of each application from the model output prior to determining contiguous regions of like classifications, and prior to determining individual crossings.

Furthermore, as the applications were non-continuous in time, it was possible for the final region classification to conflict with the initial classification for consecutive applications. For example, an application to a pair of inbound crossing intervals: BS_IN and MP_IN, would typically end in magnetosphere classifications, matching the starting classifications for the following application: MP_OUT and BS_OUT. However, if instead the application to the inbound intervals ended with another classification, such as magnetosheath, this would not match the following application and the overall output would be discontinuous. Treating the crossing intervals as ground

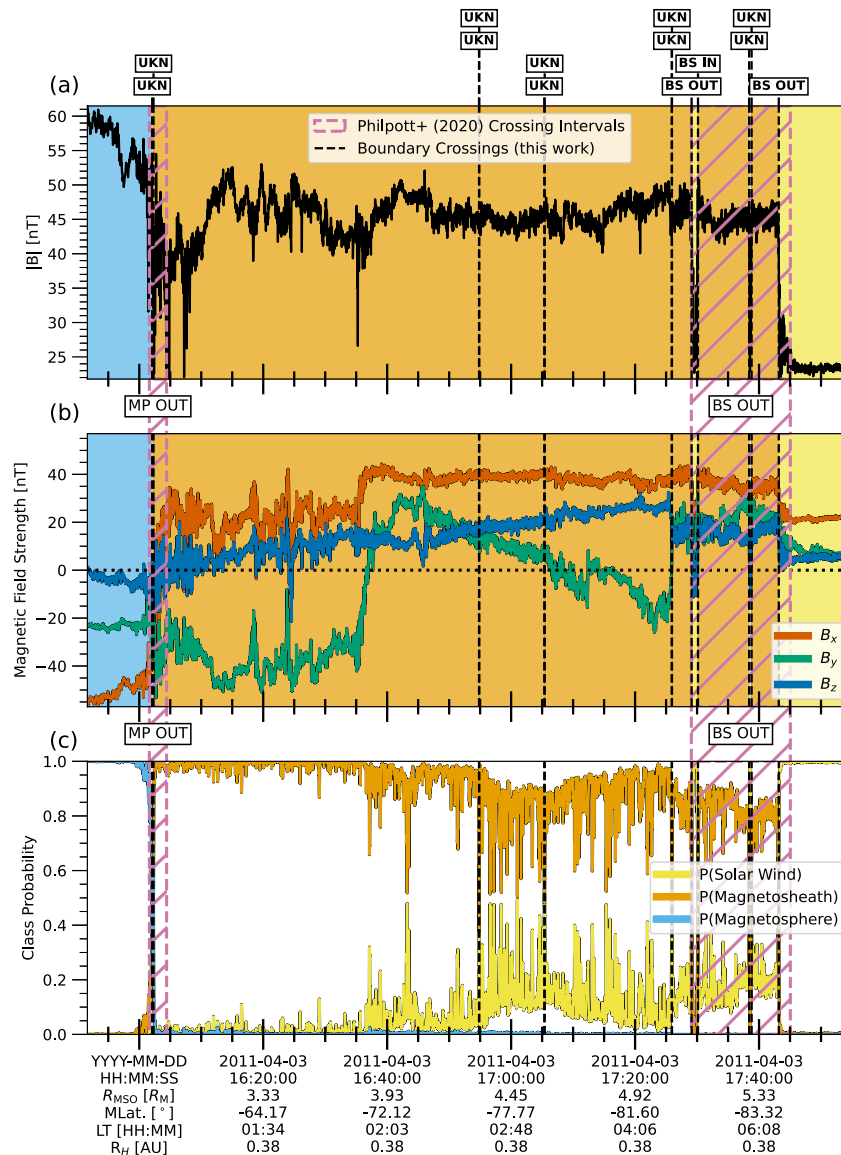


Figure 8. An example of the model application to an outbound section of MESSENGER MAG data from 3rd April 2011, 16:02 to 17:55. Panels (a, b) show the magnitude and components respectively of the 1 s averaged magnetic field as measured by MAG. Panel (c) shows the model output probabilities for each class: solar wind (P(SW), yellow), magnetosheath (P(MSh), orange), and magnetosphere (P(MSp), blue). Unknown regions (UKN) are shaded gray. The coloring of the shaded background denotes classifications, with color corresponding to the classes in panel (c). Pink diagonal hatching indicates boundary crossing intervals by Philpott et al. (2020). Individual boundary crossings are marked by black vertical dashed lines, and labeled at the top of the figure.

truth, we compared the starting and ending region predictions of each application—after determining contiguous region classifications—to what the expected classification was. If one or both of these bookending regions were misclassified, we changed the region classification to match what was expected. This step was strongly physically motivated, and affected only 76 cases (~1%).

Additionally, due to the introduction of unknown regions, it was possible for region transitions between confident regions to occur through an unknown region—that is $MSp \rightarrow UKN \rightarrow MSh$. In these cases, we have only counted unknown crossings in Table 3, and not a direct crossing—such as MP_OUT . This resulted in some crossing intervals having no direct crossings potentially skewing any statistics surrounding direct crossings only. Given that the scientific focus of our modeling approach is to produce an unambiguous and continuous list of crossings, we apply one further step to our model output: In cases where a known region transition occurs through an unknown region we placed a crossing that represents the region transition between the regions neighboring the unknown region at the midpoint of the unknown region.

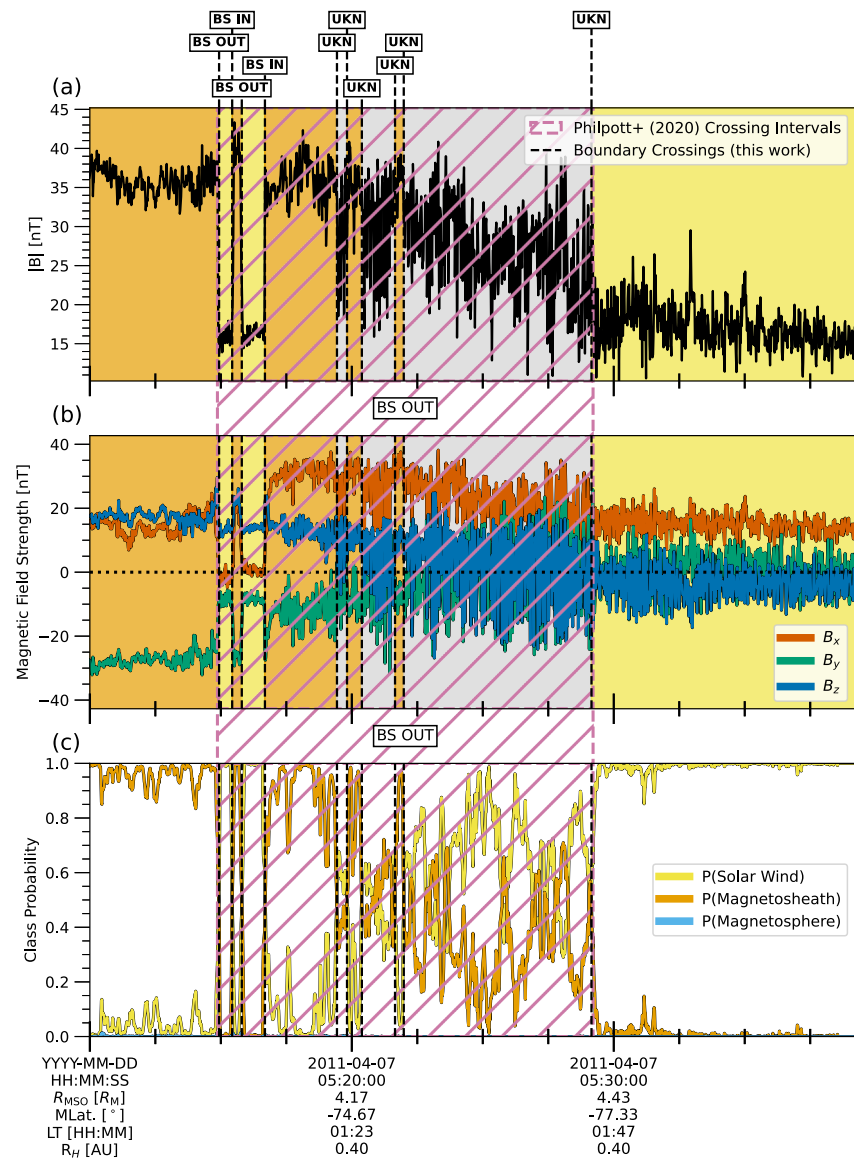


Figure 9. An example of the model application to an outbound section of MESSENGER MAG data from 7 April 2011, 05:00 to 05:39. Three unknown regions were placed at times where the model probabilities (panel (c)) were indistinct and changed prediction frequently. Figure formatting follows that described in Figure 8.

In the remainder of this study we focus on the output of the model after the post-processing steps outlined above, however in Hollman et al. (2025b) we provide both the raw output and the post-processed outputs. Additionally, we updated Table 3 with the crossing counts after the post-processing described in this section.

4. Results

4.1. Model Performance

In total we extracted 243,234 individual 10 s samples from the MESSENGER MAG data, with 218,911 (90%) of these dedicated to training the model, and 24,323 (10%) used for testing. Figure 11 shows feature importance (determined by mean decrease impurity (Ishwaran, 2015), implemented in the Scikit-learn classifier) along with a confusion matrix of classifications on the testing data set. It is important to distinguish between the accuracy of the model and the accuracy of the application. Model accuracy was directly measurable as the fraction of correctly classified samples from the testing data set. This model achieves an accuracy score of 0.9853 ± 0.0002 on the testing data set, however the accuracy score may not reflect the classification performance on data within the

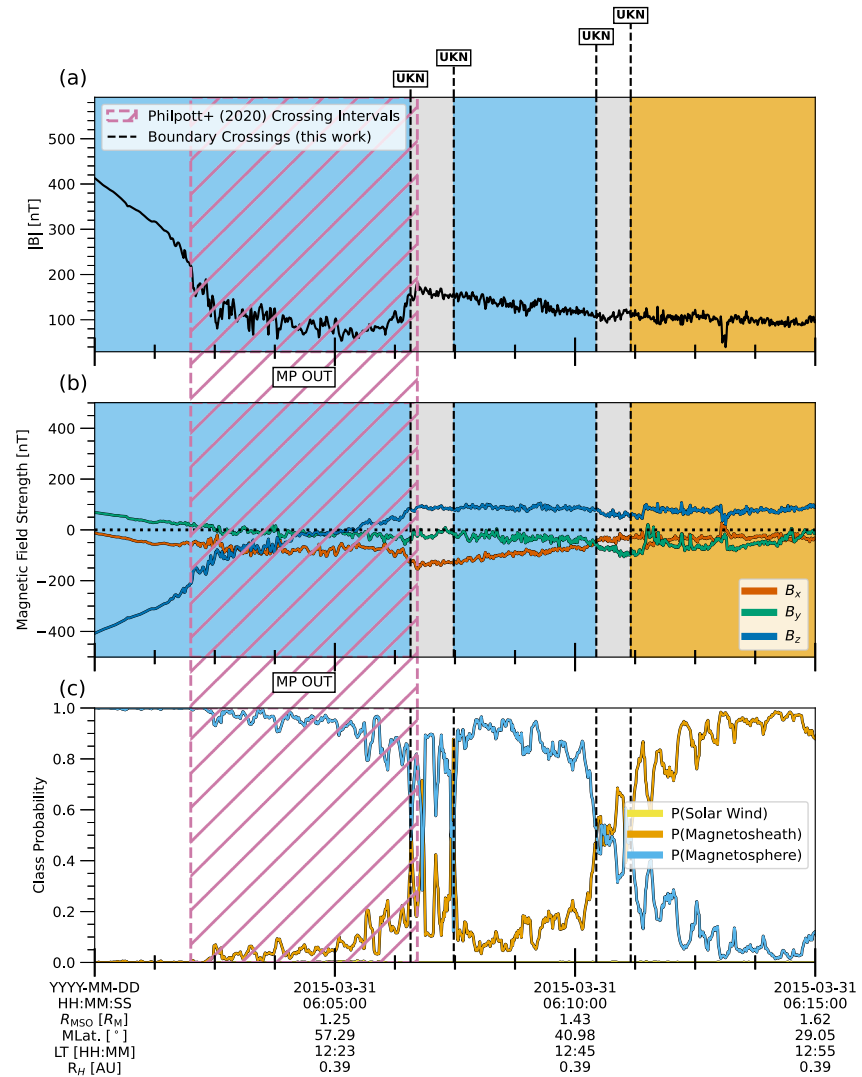


Figure 10. An example model application where model predictions visually diverge from expectations. Panel (a) shows MESSENGER MAG data from 31 March 2015, 06:00 to 06:15. Figure formatting follows that described in Figure 8. The model predicts magnetosphere (blue) before, during, and after this interval, and only predicts magnetosheath ~5 min after the crossing interval ends. Figure formatting follows that described in Figure 8.

boundary crossing intervals, as we lack ground truth labels for these regions. Accuracy on the training data set and OOB score were $\sim 1 \pm \sim 0$ and 0.9869 ± 0.0001 respectively. We note that while the training accuracy implies overfitting to the data, given the size of the training and testing data sets and the performance of the model on the unseen data, the model is clearly able to generalize well.

After the introduction of unknown regions as described in Section 3.3, we mapped the spatial distribution of all region classifications in Figure 12. Each panel shows the fraction of regions of a given classification in spatial bins as projected to the cylindrical plane, with Mercury and the average bow shock and magnetopause boundary locations drawn in black. We observed an increase in unknown regions around these average boundary locations. This was expected, as the applications of the model occurred solely around boundary crossing intervals. We also see an increase in unknown classifications far from Mercury, where more extreme events were required for crossings (and hence the previously labeled crossing intervals) to occur there.

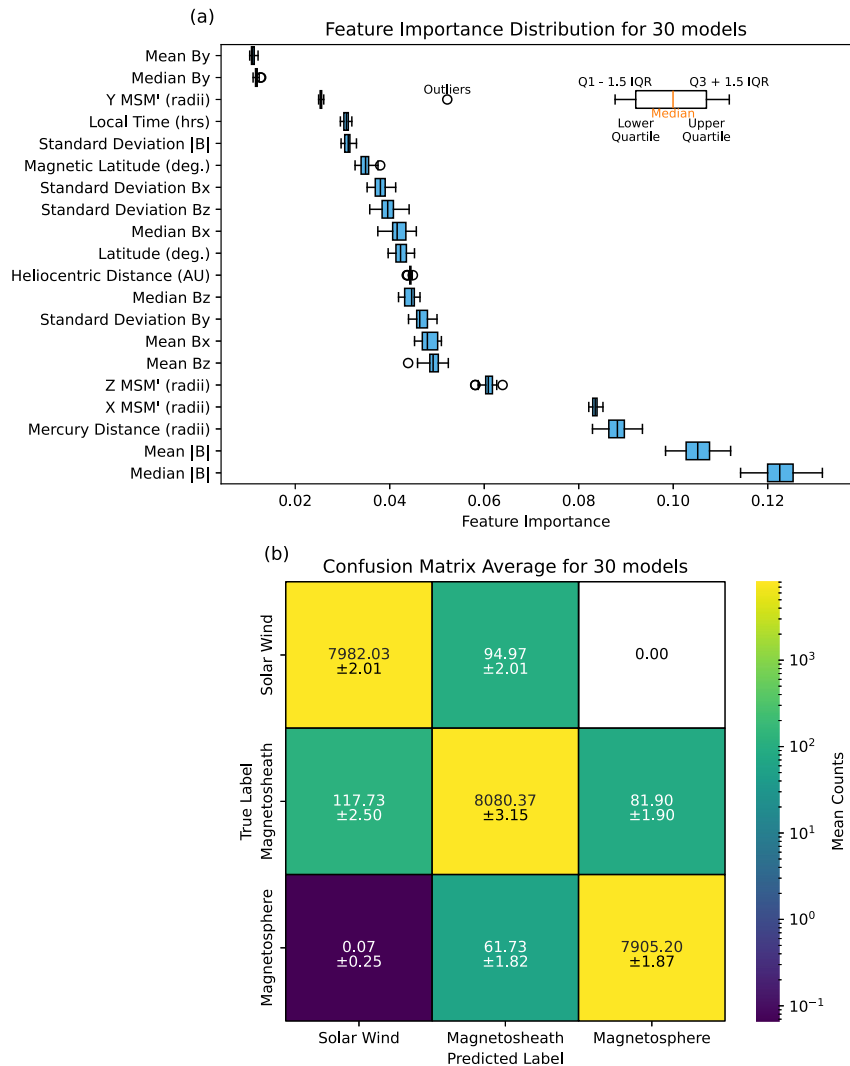


Figure 11. Average feature importance and confusion matrix for 30 random forest models with the optimized hyperparameters. Each model was initialized with a different, but fixed, random seed so that models could vary during training despite acting on the same data set. We determined impurity-based feature importance for each model and plotted their distributions in panel (a). Box and whisker plots show the distribution of importances across models as described in the key at the top right. The interquartile range (IQR) is marked with a blue box, with a black vertical line marking the median. Whiskers extend out to 1.5 times the IQR before and beyond the 1st and 3rd quarters respectively. Outliers to these are marked by circles. Panel (b) shows the mean and standard deviation of confusion matrices calculated on the testing data set, colored by number of classifications on a log scale. Correct classifications exist on the diagonal of this matrix.

4.2. Newly Identified Boundary Crossings

We identified a total of 100,901 new boundary crossings across the MESSENGER mission—an average of ~ 5 individual boundary crossings per crossing interval. Table 3 shows counts for each crossing type. This method finds 19,553 direct boundary crossings and 81,348 unknown crossings. The imbalance between direct and unknown crossings exists as unknown regions were determined based on region duration, along with region confidence. Shorter duration regions were more likely to be considered unknown (as shown in Figure 7) and hence the average number of unknown regions in a given timespan, and therefore also of unknown crossings, was larger than that for direct crossings. As discussed in Section 3.2, direct crossings may, but don't necessarily, occur within unknown regions, though this method lacks the accuracy required to determine them. In addition to direct crossings, indirect (but definite) crossings also occur, where one region transitions to another through an unknown region (i.e., solar wind $\rightarrow$ unknown $\rightarrow$ magnetosheath). In these cases, we record two unknown crossings, though physically, at least

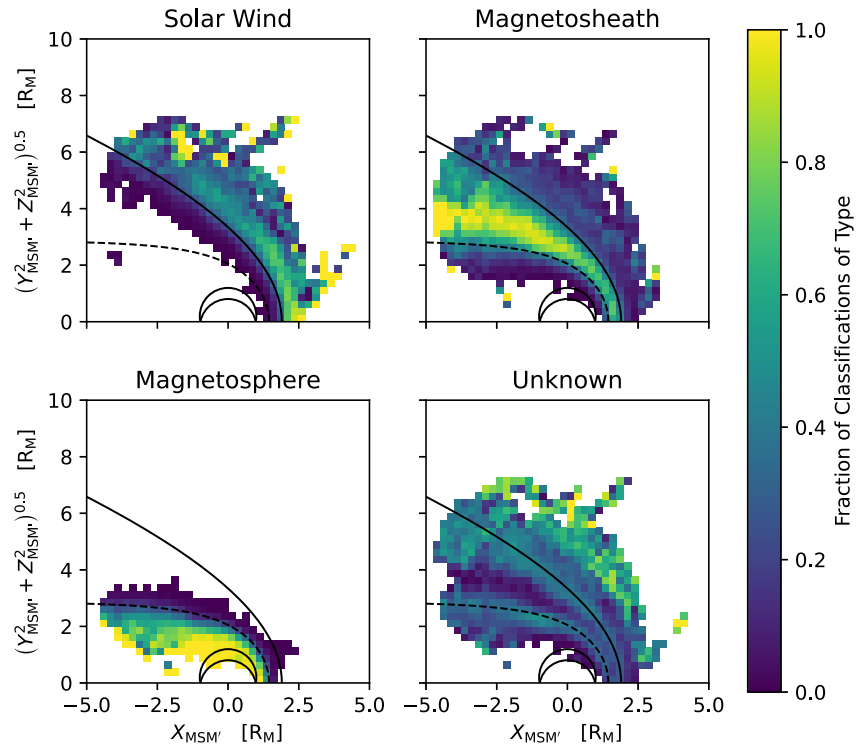


Figure 12. The spatial distribution of solar wind, magnetosheath, magnetosphere, and unknown classifications as projected to the cylindrical plane in the aberrated Mercury-Solar-Magnetospheric (MSM') coordinate system. Two overlapping circles were drawn to represent the northern and southern extents of Mercury, folded into the plane. The average boundary shape by Winslow et al. (2013) for the bow shock and magnetopause are drawn as a solid and dashed curve respectively.

one—but always an odd number—of direct crossings must have occurred. This issue is partially accounted for by post processing steps described in Section 3.4, with updated counts for each crossing type recorded in Table 3.

After post-processing, we found a total of 31,408 crossings. We plotted the spatial distribution of these post-processed crossings in Figure 13 which visually compare well to those of the boundary crossing intervals in Figure 4. However, we leave further statistical analysis for future works. This list of post-processed crossings was made publicly available for use (Hollman et al., 2025b).

5. Discussion and Future Work

5.1. Evaluating Unknown Regions

As described in Section 3.4, within the post-processing steps of this work we defined some contiguous sections sharing the same classification label as unknown regions under the condition that they were shorter than a threshold on duration, and less confident than a threshold on region confidence. It is worthwhile discussing the interpretation of these unknown regions. They were constructed empirically, and hence we should be cautious when considering their meaning. We know the model has generalized well to the physics of this system. Its performance on the testing data set has shown that it can separate solar wind, magnetosheath, and magnetosphere data with very high accuracy. By definition, these unknown regions occur where the model lacks “confidence”, at times where the ensemble is split in their classification, and is quite variable in subsequent time-steps. However, there are times when this can be informative. Consider Figure 9: before and after the outbound bow shock interval, we see distinct differences between magnetosheath and solar wind. The unknown region (highlighted in gray) contains data which differs significantly from both the magnetosheath before, and solar wind after. In this case, the model informs us that these times are dissimilar to its training data set, and we can interpret this physically as processes at the boundary, for example introducing effects such as turbulence, resulting in data which was physically dissimilar from that which the model trained on. This is furthered by Figure 12, which plots the proportion of each region (solar wind, magnetosheath, and unknown) on a cylindrical spatial grid. We found

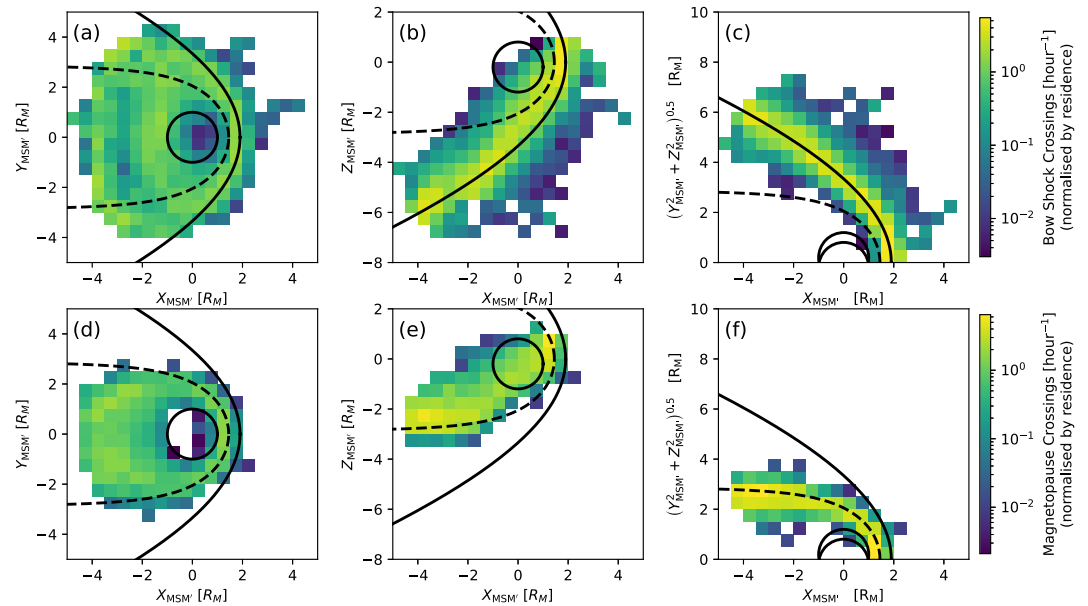


Figure 13. The spatial distribution of individual bow shock (panels (a–c), above) and magnetopause (panels (d–f), below) crossings as determined based on changes in region probability are shown as projected to the XY and XZ, and cylindrical planes of the aberrated Mercury-Solar-Magnetospheric (MSM') coordinate system. A circle was drawn to represent the size and location of Mercury. In the case of the cylindrical planes two overlapping circles were drawn to represent the northern and southern extents of Mercury along the Z axis, folded into the plane. The average boundary shape by Winslow et al. (2013) for the bow shock and magnetopause are drawn as a solid and dashed curve respectively.

unknown regions to be gathered near the average boundary determined by Winslow et al. (2013), where we would expect to see such near-boundary effects most commonly. Additionally, we see an increase in unknown regions with distance from the average bow shock location, where extreme conditions are required for application in these bins. We cannot however completely rely on the introduction of unknown regions to inform us of physical discrepancies as we also find cases where unknown regions occur without an obvious difference in data, for example, the brief unknown regions between the MP OUT and BS OUT crossing intervals in Figure 8.

It is important to highlight the quantity of unknown crossings determined compared to that of direct crossings. Prior to the removal of these unknown crossings in post-processing steps, the ratio of unknown crossings to direct crossings was roughly 3:1. This appears to directly conflict with the high accuracy of the model on the testing data set as, if all classifications were highly confident, we would solely have direct crossings. As discussed in Section 3.3, unknown regions were by definition shorter on average than any of the known classifications, and hence unknown crossings must occur more frequently than direct crossings, resulting in an increased count.

The post-processing steps introduced make an effort to replace these unknown crossings with direct crossings where applicable. In particular, we would like to highlight and contrast the first unknown region in Figure 8 (separating magnetosphere and magnetosheath), and the last (and largest) unknown region in Figure 9 (separating magnetosheath and solar wind). In both of these cases we require at least one physical crossing to have occurred. The post-processing steps replace the two unknown crossings bookending this region with a single direct crossing (labeled based on the region classifications before and after) placed in the middle of this region. For the example in Figure 8, this method works well as this region is short, and the placement is temporally accurate. However, for the example in Figure 9, the large length of this unknown region means that placement in the middle results in a large margin of uncertainty around the exact location of the crossing. To eye, the true placement of the bow shock crossing within this unknown region is at the end, and hence, the placement of this crossing after the post-processing steps was offset by roughly 4 min. We can infer from the distribution of unknown regions that there is an average uncertainty where unknown regions are present (Median: ± 13 s, Q1 = ± 4 s, Q3 = ± 40 s, 95th percentile: ± 2 min 36 s). For this reason, the post-processed version of this list is most useful for large scale statistical studies, however for event studies, the raw model output is a better reference and should be evaluated manually by the user.

5.2. Future Work

This work opens numerous possibilities for further investigation. We present a random forest model as a method for magnetospheric region classification with magnetic field and ephemeris data alone. As the model accuracy was high, the possible percentage gains in improving the model are low, however further tuning of hyperparameters is possible, and exploration of additional features could still prove useful. The set of features explored was not exhaustive. Additionally, further simplification of the model could be explored by removing low importance features. Both the tuning of hyperparameters, and the further investigation of features could improve model performance in testing and application.

We have applied a new, generalizable method to detect Mercury boundary crossings. This manuscript, along with our open source code (Hollman et al., 2025a) will be a strong foundation for boundary detection for data from the future BepiColombo mission. This dual-spacecraft mission will be the first to offer multiple-viewpoint in situ observations at Mercury. The Mercury Magnetospheric Orbiter (MMO) will enter into a highly eccentric orbit of Mercury, sampling from the solar wind through to the magnetosphere, while the Mercury Planetary Orbiter (MPO) will enter a relatively circular orbit sampling primarily the magnetosphere and magnetosheath (Benkhoff et al., 2021). Magnetospheric region classification and boundary detection will be vitally important for the coordination of science between both spacecraft. The two spacecraft will follow different orbits than MESSENGER, and while they may at times overlap with orbits by MESSENGER, they will sample a broader range of positions. As a result, ephemeris features extracted from BepiColombo when directly applying this model may fall outside the ranges observed during training, and affect model performance. To circumvent this, we performed a preliminary investigation into the performance of the model when excluding ephemeris features. This model performed similarly to the original model still achieving an average accuracy of 0.9190 ± 0.0004 . A detailed description and comparison can be found in Appendix A. We hope to test both models with/without ephemeris features without any transfer learning on BepiColombo data in the future, however further labeling may be required to adapt this method to those spacecraft. Additionally, while the model used in this study is Mercury specific in terms of feature values, the method is entirely generalizable to any magnetized planet in the solar system (provided some labeling of magnetospheric regions has been carried out).

This work also paves the way for a wealth of new statistical studies. Previous statistical studies on the location of Mercury's magnetospheric boundaries have been based off lists of crossing intervals, specifically taking the center of the interval as the boundary location (He et al., 2022; Philpott et al., 2020; Winslow et al., 2013). However, with knowledge of the precise times (and locations) of each boundary crossing, as presented in this work, we can accurately connect perturbations of the boundary location to external and internal drivers (solar wind dynamic pressure, Alfvén Mach number, reconnection rates, core conduction, etc.). Additionally, we can unravel variability in the boundary shape, as crossing intervals contain a differing number of individual boundary crossings. Furthermore, the number of crossings for a given pass of each boundary gives an indication of boundary variability and could be explored further to determine differences in variability from the nose the flanks of the magnetosphere, along with differences in variability due to Mercury's heliocentric distance. To aid in the use of the resulting raw model output and post-process crossing lists in future works, we have included a detailed guide along with the software repository (Hollman et al., 2025a).

6. Conclusions

We present a new list of crossings by the MESSENGER spacecraft of Mercury's bow shock and magnetopause based on automated region classification. A random forest was fit to MESSENGER magnetic field and ephemeris data to classify regions of solar wind, magnetosheath, and magnetosphere. The random forest is highly accurate on the testing data set, with a score of 0.9853 ± 0.0002 , and its performance was visually discussed over three example applications. Several post-processing steps were introduced to create an unambiguous and continuous list of bow shock and magnetopause crossings for the duration of the MESSENGER mission. Furthermore, our model and method are generalized and applicable to the future BepiColombo mission, and other magnetized planets in the solar system. We highlight how models trained on MESSENGER data have direct application to future data from BepiColombo.

Appendix A: Evaluating a Model Excluding Ephemeris

The random forest model used in this study makes predictions based on features, some of which are derived from MESSENGER's trajectory (position, local time, latitude, etc.). These features are useful in achieving a high accuracy, with X_{MSM} , Z_{MSM} , and distance to Mercury being in the top 5 most important features (Figure 11). However, they restrict successful application of the model to the parameter-space which MESSENGER explored. Future spacecraft at Mercury, such as those from the BepiColombo mission, will orbit on different trajectories to MESSENGER's and will take measurements in positions not sampled before. In these instances, a model which includes ephemeris features could produce inaccurate results. In order to showcase the generalizability of this method, we highlight how a version of this model retains a high accuracy when ephemerides are excluded. A random forest model was fit to the same training data set as described in 2.3 with the exception that ephemeris features were removed. The same method for optimisation and performance evaluation was used. This model had reduced accuracy in comparison with the previous, as one would expect, but remains highly accurate with a score of 0.9190 ± 0.0004 . Figures A1a and A1b show the feature importance and confusion matrix respectively.

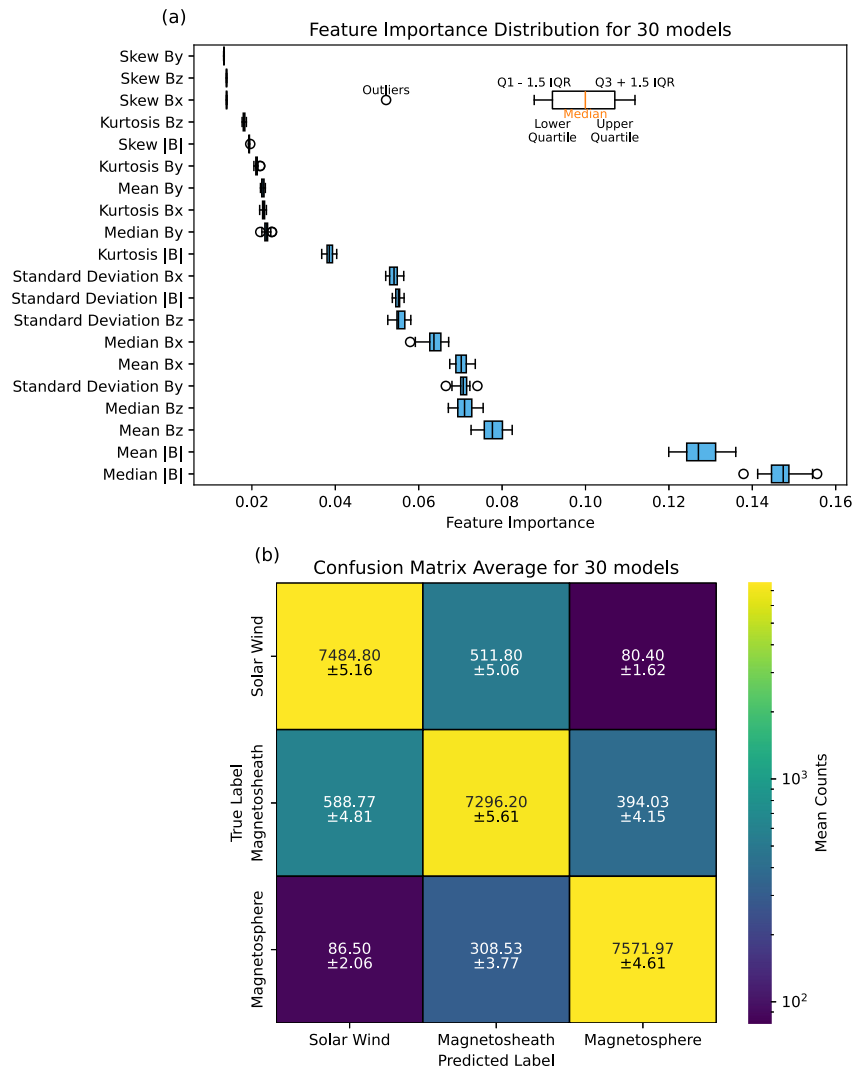


Figure A1. Average feature importance and confusion matrix as described in Figure 11 for a model retrained without the inclusion of ephemeris features.

Appendix B: Crossing Intervals Containing Unexpected Classifications

As described in Section 3.3, our model made predictions of solar wind within the magnetopause crossing intervals previously labeled by Philpott et al. (2020). There were 13 of these intervals, which are detailed in Table B1.

Table B1

Interval label	Orbit number	Interval start time (YYYY-MM-DD HH:MM:SS)	Interval end time (YYYY-MM-DD HH:MM:SS)	Duration of solar wind Classifications(s) [% of interval]	
MP_OUT	28	2011-04-01 03:50:33	2011-04-01 04:11:11	275	(22.2)
MP_OUT	472	2011-11-07 22:33:40	2011-11-07 22:44:14	5	(0.7)
MP_IN	816	2012-04-22 13:48:22	2012-04-22 14:04:50	925	(93.6)
MP_OUT	817	2012-04-22 23:48:06	2012-04-23 00:13:15	566	(37.5)
MP_IN	868	2012-05-09 20:41:00	2012-05-09 21:01:25	197	(16.1)
MP_OUT	927	2012-05-29 15:48:26	2012-05-29 15:58:20	101	(17)
MP_OUT	931	2012-05-30 23:48:25	2012-05-31 00:05:17	7	(0.7)
MP_OUT	1,020	2012-06-29 16:26:30	2012-06-29 17:17:50	805	(26.2)
MP_IN	1,241	2012-09-11 07:09:34	2012-09-11 07:11:08	18	(19.1)
MP_IN	1,273	2012-09-21 22:47:23	2012-09-22 22:54:29	329	(77.1)
MP_OUT	1,273	2012-09-22 01:26:12	2012-09-22 01:37:22	506	(75.6)
MP_OUT	1,936	2013-04-30 22:09:18	2013-04-30 22:09:18	21	(1.2)
MP_IN	2,536	2013-11-17 00:54:41	2013-11-17 01:02:52	32	(6.5)
MP_OUT	2,710	2014-01-14 02:19:10	2014-01-14 02:28:38	64	(11.3)
MP_IN	2,816	2014-02-18 09:38:01	2014-02-18 09:40:51	12	(7.1)

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

MESSENGER data are available through the Planetary Plasma Interactions Node of the Planetary Data System (Korth & Anderson, 2016). The training data set, raw model output, and post-processed crossing list created by this work are available for download (Hollman et al., 2025b). All code for data set construction, modelling, application, and figure creation for this work are publicly available (Hollman et al., 2025a).

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